

Executive Summary

E-Commerce Sales & Customer Behaviour Analysis

This project focused on analysing transactional sales data from an e-commerce company to identify business trends, profitability drivers, and customer behaviour patterns. The dataset initially contained missing values, inconsistencies, duplicate records, and logical errors, which were systematically cleaned and validated before conducting detailed exploratory data analysis (EDA).

Key Findings

- **North region generated the highest sales**, making it the strongest performing region.
- The **Electronics category recorded the highest total sales**, indicating strong customer demand.
- A **strong positive correlation (0.87)** was observed between Sales and Profit, meaning increased sales directly drive higher profitability.
- **Discount shows a negative relationship with Profit**, indicating that excessive discounting reduces profit margins.
- The **Consumer segment generated the highest total revenue**, while the **Corporate segment had the highest average order value (₹52,193)**.
- Pareto analysis revealed that the top 20% of customers contribute only **29% of total revenue**, indicating revenue is distributed across a broad customer base rather than concentrated among a few customers.
- Shipping cost showed minimal impact on overall profitability.

Business Implications

The analysis indicates that business growth is primarily driven by increasing sales volume rather than relying heavily on discount strategies. Revenue stability is supported by a diversified customer base, reducing dependency risk.

Strategic Recommendations

1. Expand marketing and operational focus in the North region.
2. Promote the Electronics category to further increase revenue.
3. Optimize discount strategies to protect profit margins.
4. Strengthen relationships with Corporate customers due to higher order value.
5. Continue broad customer acquisition efforts to sustain balanced revenue growth.

Conclusion

Overall, the analysis provides data-driven insights into regional performance, product category contribution, customer behaviour, and profitability drivers. These findings can support strategic decision-making to enhance revenue growth and operational efficiency.